

THE CHOW RANK OF THE FIVE-BY-FIVE PERMANENT

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ABSTRACT. Let perm_5 denote the permanent of a generic 5 by 5 matrix. Over every field of characteristic zero, we prove that its ordinary Chow rank is 16. This certificate-driven presentation separates the written geometric argument from four finite computational lemmas. Their inputs, exhaustive domains, arithmetic rings, expected outputs, and failure conditions are stated in the article. The largest endpoint exhausts 886,464 coordinate flags over each of $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ and checks every extremum over $\mathbb{Q}$; independent programs also verify Glynn's identity and all $\binom{25}{5}$ coordinate five-planes. A deterministic fail-closed replay links these certificates to the written proof.

1. INTRODUCTION

Notation and conventions. Throughout the paper, $\mathbb{K}$ denotes a field of characteristic zero. For a positive integer n , write $[n] = \{0, 1, \dots, n-1\}$, let S_n be the group of bijections of $[n]$, and let $\binom{\Omega}{a}$ denote the set of all a -element subsets of a finite set Ω . If W is a finite-dimensional $\mathbb{K}$ -vector space, then W^* , $\text{Sym}^a W$, $\Lambda^a W$, $\text{Gr}(a, W)$, and $\mathbb{P}(W)$ denote its dual, its degree- a symmetric and exterior powers, the Grassmannian of a -dimensional subspaces, and the space of one-dimensional quotients, respectively. For $L \subset W$, the annihilator is $L^\perp = \{\lambda \in W^* : \lambda|_L = 0\}$. We use span , im , ker , rank , dim , and codim with their usual linear-algebraic meanings. For a prime p , $\mathbb{F}_p$ is the field with p elements.

Let

$$V = \text{span}_{\mathbb{K}}\{x_{ij} : (i, j) \in [5] \times [5]\}, \quad X = (x_{ij})_{i,j \in [5]}.$$

For subsets $I, J \subset [5]$ of the same cardinality, $X_{I,J}$ is the submatrix of X with row set I and column set J , and $\text{perm}(X_{I,J})$ is its permanent, defined by the same permutation formula as below. Define

$$\text{perm}_5 = \sum_{\sigma \in S_5} \prod_{i=0}^4 x_{i, \sigma(i)} \in \text{Sym}^5 V.$$

For $f \in \text{Sym}^m V$, $0 \leq a \leq m$, and $\lambda_1, \dots, \lambda_a \in V^*$, let $\partial_{\lambda_1} \cdots \partial_{\lambda_a} f$ denote iterated directional differentiation. The order- a catalecticant and the

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remaining degree- j derivative space, where $j = m - a$, are

$$\begin{aligned} C_{a,j}(f) &: \text{Sym}^a V^* \longrightarrow \text{Sym}^j V, \\ C_{a,j}(f)(\lambda_1 \cdots \lambda_a) &:= \partial_{\lambda_1} \cdots \partial_{\lambda_a} f, \\ \mathcal{D}_j(f) &:= \text{im } C_{a,j}(f). \end{aligned}$$

For a subspace $G \subset \text{Sym}^j V$, we write $\partial G = \text{span}\{\partial_\lambda g : \lambda \in V^*, g \in G\}$. If $q \in \text{Sym}^2 V$, then $M(q)$ is the symmetric matrix of the associated bilinear form in the ordered basis (x_{ij}) ; the rank of q means $\text{rank } M(q)$.

A quintic Chow term is a nonzero product of five members of V , regarded as linear forms on V^* . The ordinary Chow rank $\text{ChowRank}_{\mathbb{K}}(f)$ of $f \in \text{Sym}^5 V$ is the least number of such terms whose sum is f ; we set $\text{ChowRank}_{\mathbb{K}}(0) = 0$. Within this paper we abbreviate it to $\text{ChowRank}(f)$. This is also called product rank in the literature; see [3].

Theorem 1.1 (Main theorem). *Over the field $\mathbb{K}$,*

$$\boxed{\text{ChowRank}_{\mathbb{K}}(\text{perm}_5) = 16.}$$

The proof is computer-assisted in a deliberately narrow sense. The geometric reductions, rank inequalities, degeneration arguments, and the passage from a hypothetical decomposition to a finite state space are written mathematical proofs. Exact programs verify finite endpoint tables obtained from those reductions. No random experiment, floating-point computation, SAT certificate, Groebner-basis computation, or unpublished data is used.

The lower-bound argument has four stages. First, a Koszul flattening has rank 2400 on perm_5 , whereas one Chow term contributes at most 240. Second, a low-rank classification gives a one-dimensional bound for the intersection of a Chow derivative space with the permanent quadratic derivative space. A projective incidence argument then reduces the unique one-intersection endpoint to coordinate flags. Third, six terms are fixed in a hypothetical decomposition of length at most fifteen. Petersen-product shadows and a double-quotient rank inequality leave exactly 58 integer states. Fourth, exact structural and finite arguments exclude all 58 states; the terminal numerical contradiction is $2215 > 9 \cdot 245 = 2205$.

Section 2 develops the invariant linear algebra. Section 3 proves the one-intersection theorem. Sections 4 and 5 carry out the fixed-six reduction and its terminal exclusions. Section 6 gives the upper bound, and Section 8 states the exact replay boundary.

2. DERIVATIVE SPACES AND KOSZUL FLATTENINGS

Write $V = A \otimes B$, where $A = \text{span}_{\mathbb{K}}\{a_0, \dots, a_4\}$, $B = \text{span}_{\mathbb{K}}\{b_0, \dots, b_4\}$, and $x_{ij} = a_i \otimes b_j$. Put

$$E = \mathcal{D}_2(\text{perm}_5) = \text{im } C_{3,2}(\text{perm}_5), \quad D = \mathcal{D}_3(\text{perm}_5) = \text{im } C_{2,3}(\text{perm}_5).$$

Their standard bases are

$$E = \text{span}\{x_{ia}x_{jb} + x_{ib}x_{ja} : i < j, a < b\}, \quad (1)$$

$$D = \text{span}\{\text{perm}(X_{I,J}) : |I| = |J| = 3\}. \quad (2)$$

Distinct basis elements have distinct monomial supports, so

$$\dim E = \binom{5}{2}^2 = 100, \quad \dim D = \binom{5}{3}^2 = 100.$$

For $F \subset \text{Sym}^2 V$, define the first prolongation

$$F^{(1)} = \{g \in \text{Sym}^3 V : \partial_\lambda g \in F \text{ for all } \lambda \in V^*\}.$$

The Koszul differential

$$\delta_F : V \otimes F \longrightarrow \Lambda^2 V \otimes V, \quad w \otimes uv \longmapsto u \wedge w \otimes v + v \wedge w \otimes u$$

has kernel naturally isomorphic to $F^{(1)}$. On a fixed set of three rows and three columns, transpositions connect the six perfect matchings. For $f \in \text{Sym}^5 V$, define its quadratic Koszul flattening to be $K(f) = \delta_{\mathcal{D}_2(f)}$. Consequently $E^{(1)} = D$, and therefore

$$\text{rank } K(\text{perm}_5) = 25 \dim E - \dim E^{(1)} = 2400. \quad (3)$$

If $T = \ell_1 \cdots \ell_5$ is a Chow term, its quadratic derivative space $F_T := \mathcal{D}_2(T)$ is spanned by the ten two-factor products, and its first prolongation contains the ten three-factor products. Upper semicontinuity under degeneration gives

$$\dim F_T \leq 10, \quad \text{rank } K(T) \leq 25 \cdot 10 - 10 = 240. \quad (4)$$

2.1. Quadrics of rank at most five.

Lemma 2.1 (Low-rank classification). *If $0 \neq q \in E$ and $\text{rank } M(q) \leq 5$, then*

$$M(q) = U \otimes W, \quad \text{rank } U = \text{rank } W = 2.$$

Here U and W are symmetric 5 by 5 matrices, and $\otimes$ denotes their Kronecker product. In particular, $\text{rank } M(q) = 4$.

Proof. Arrange $M(q)$ as a 5 by 5 block matrix (D_{ij}) by rows of variables. Each D_{ij} is symmetric with zero diagonal, and $D_{ii} = 0$. Choose a nonzero block B_0 . The principal block $\begin{bmatrix} 0 & B_0 \\ B_0 & 0 \end{bmatrix}$ has rank $2 \text{rank } B_0$. A nonzero symmetric zero-diagonal matrix cannot have rank one, so $\text{rank } B_0 = 2$. Any connecting block with nonzero image in the cokernel of this principal block would increase the total rank by at least two, contrary to the rank-five hypothesis. Hence every connecting block has image in $\text{im } B_0$.

A rank-two symmetric zero-diagonal matrix is $ab^\top + ba^\top$, with disjoint supports for a and b . Inside the fixed image $\text{span}\{a, b\}$, the symmetric zero-diagonal condition forces every block to be a scalar multiple of B_0 . Congruence Schur elimination on the first two block rows leaves a symmetric zero-diagonal complement of rank at most one, hence the complement is zero.

Thus every block is proportional to B_0 , so $M(q) = U \otimes B_0$. Multiplicativity of rank and the zero-diagonal condition give $\text{rank } U = \text{rank } B_0 = 2$; take $W = B_0$. $\square$

Corollary 2.2 (One-dimensional intersection). *If $L \subset V$ and $\dim L \leq 5$, then*

$$\dim(E \cap \text{Sym}^2 L) \leq 1.$$

Consequently $\dim(E \cap F_T) \leq 1$ for every quintic Chow term T .

Proof. If the intersection contained a projective line, Lemma 2.1 would place that line in a Segre variety of rank-two tensor products. A line contained in a Segre variety fixes one tensor factor. Two nonproportional choices for the other factor have images spanning dimension at least three; the essential-variable space then has dimension at least $2 \cdot 3 = 6$, a contradiction. $\square$

3. THE ONE-INTERSECTION FLAG THEOREM

Let $Q = \text{Sym}^2 V/E$, and let $\pi : \text{Sym}^2 V \rightarrow Q$ be the quotient map. For a subspace $W \subset Q$, define its relative prolongation excess by

$$p(W) = \dim(\pi^{-1}(W)^{(1)}) - \dim D.$$

Thus no choice of a lift of W is involved.

Theorem 3.1 (One-intersection flag theorem). *Let $T = \ell_1 \cdots \ell_5$ be a Chow term satisfying*

$$\dim F_T = 10, \quad \dim(E \cap F_T) = 1.$$

If $Z := \pi(F_T)$ is the nine-dimensional image of F_T in Q , and

$$Z \subset W \subset Q, \quad \dim W = 10,$$

then $p(W) \leq 26$.

Proof. Extend scalars to an algebraic closure. Put $L_T = \text{span}\{\ell_1, \dots, \ell_5\}$, with $d = \dim L_T \in \{4, 5\}$. Consider the incidence variety in

$$\text{Gr}(d, V) \times \text{Gr}(10, \text{Sym}^2 V) \times \text{Gr}(9, Q) \times \text{Gr}(10, Q) \times \mathbb{P}(E)$$

whose points $(L, F, Z', W', [q])$ satisfy

$$F \subset \text{Sym}^2 L, \quad q \in F, \quad F \subset \pi^{-1}(Z'), \quad Z' \subset W'. \quad (5)$$

These are zero loci of maps of universal bundles; the incidence variety is therefore projective and stable under the row-column diagonal torus. The orbit closure of the point defined by T contains a torus-fixed point by the Borel fixed-point theorem [1, Chapter I, Theorem 4.7]. Corollary 2.2 preserves the one-dimensional intersection at the limit, while upper semicontinuity of kernel dimension shows that it suffices to bound the relative prolongation at the fixed point.

The quotient Q is multiplicity-free for the row-column diagonal torus. Indeed, its one-dimensional weight spaces consist of the (25) square weights, the (50) same-row weights, the (50) same-column weights, and one quotient

line for each of the (100) rectangles: in a rectangle the two matching monomials have the same weight and their sum spans the corresponding line of (E). Thus

$$\dim Q = 25 + 50 + 50 + 100 = 225,$$

and all these quotient weights are distinct. Consequently, at a fixed point, L, F, Z', W' , and $[q]$ are coordinate objects. The rank-four form q is a rectangle permanent. The finite endpoint is therefore a completely specified enumeration of coordinate four- and five-plane flags. For each flag the program constructs the divided-power integer matrix of the relative prolongation map. It checks

21,600 four-plane flags,
432,432 attached five-plane flags,
432,432 external five-plane flags.

The maxima are respectively 22, 26, 22; exactly twenty attached flags attain 26. The entire enumeration is repeated over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, with identical histograms, and all twenty extrema are row-reduced over $\mathbb{Q}$. For an integer matrix M , $\text{rank}_{\mathbb{F}_p} M \leq \text{rank}_{\mathbb{Q}} M$, hence $\dim \ker_{\mathbb{Q}} M \leq \dim \ker_{\mathbb{F}_p} M$. Thus the finite-field computation is used only in the valid direction, as an upper bound on the characteristic-zero kernel. This proves $p(W) \leq 26$. $\square$

Remark 3.2. The endpoint enumeration is a premise of the proof. It is independently implemented by `scripts/perm5_one_intersection_independent_multifield.py`; it neither imports the frozen generator nor reads the frozen result.

4. THE FIXED-SIX REDUCTION

Equations (3) and (4) first give $\text{ChowRank}(\text{perm}_5) \geq 10$. Suppose, for contradiction, that a minimal decomposition has length $r \leq 15$. Then $10 \leq r \leq 15$. Keep six original nonzero terms fixed and repeatedly replace a remaining nonzero term T by $\frac{1}{2}T + \frac{1}{2}T$. This produces

$$\text{perm}_5 = T_1 + \cdots + T_6 + T_7 + \cdots + T_{15},$$

where the residual $R = T_7 + \cdots + T_{15}$ has exactly nine Chow terms. No separate lower bound of fifteen is used.

Let $C = T_1 + \cdots + T_6$, and put

$$J = \text{im } C_{2,3}(C), \quad H = \text{im } C_{3,2}(C), \quad S = D \cap J, \quad G = E \cap H,$$

$$s = \dim S, \quad t = \dim G, \quad d = \dim(H/G), \quad h = \dim H = t + d.$$

For $1 \leq i \leq 6$, set $F_i := \mathcal{D}_2(T_i)$, and write $U := F_1 + \cdots + F_6$. Only $H \subset U$ is asserted; equality is reserved for the near-maximal states proved below.

The bases of D and E are indexed by pairs of three-subsets and pairs of two-subsets. Complementation identifies the derivative shadow with the open neighborhood operator in the product of two Petersen graphs. Its exact minimum-shadow table is displayed below. The terminology follows

classical shadow theory [5, 4], although this table is a graph-specific exact computation.

q	0	1	2	3	4	5	6	7	8	9	10	11
μ_q	0	9	15	18	18	24	27	27	30	30	30	35
q	12	13	14	15	16	17	18	19	20	21	22	23
μ_q	35	36	36	36	36	42	45	45	48	48	48	52.

(6)

Here μ_q is the minimum size of the open neighborhood of a q -vertex subset in the product graph. The dynamic recurrence uses only

$$\nu(k) = (0, 3, 5, 6, 6, 8, 9, 9, 10, 10, 10), \quad 0 \leq k \leq 10,$$

where $\nu(k)$ is the corresponding minimum-neighborhood function for one Petersen graph. The recurrence is exact integer arithmetic and is reconstructed by the replay.

Let $\overline{H} := \pi(H) = (E + H)/E \subset Q$, and set $p_d := p(\overline{H}) = \dim((E + H)^{(1)}/D)$; the subscript records $d = \dim \overline{H}$. Since $J \subset (E + H)^{(1)}$, $D \cap J = S$, and $\partial S \subset G$,

$$p_d \geq t + d - s \geq \mu_s + d - s. \quad (7)$$

Applying Sylvester's rank inequality to the two quadratic Koszul maps, and using $\text{rank } K(C) \leq 24h$, gives the central double-quotient inequality

$$\text{rank } K(R) \geq 2400 + (25d - p_d) - (25s - \mu_s). \quad (8)$$

Corollary 2.2, applied after lifting $E \cap U$ to $\bigoplus_i F_i$ and projecting to five blocks, gives

$$t \leq \dim(E \cap U) \leq 5 \cdot 10 + 1 = 51. \quad (9)$$

Since $\mu_{23} = 52$, we have $s \leq 22$. The exact global prolongation bounds together with (7)–(9) exclude $s \leq 18$ and $d \leq 8$. Thus

$$19 \leq s \leq 22, \quad d \geq 9, \quad h \leq 60.$$

For fixed s , the possible (t, d) form an integer triangle of side $53 - \mu_s$. Hence their number is exactly

$$\binom{8}{2} + 3\binom{5}{2} = 58.$$

Lemma 4.1 (Near-maximal coupling). *If a fixed-six state satisfies $h \geq 57$, then $H = U$.*

Proof. Put

$$e = 60 - \sum_i \dim F_i, \quad k = \sum_i \dim F_i - \dim U, \quad c = \dim U - h.$$

Then $e + k + c = 60 - h \leq 3$. If $c > 0$ and $k = 0$, differentiation of a cubic relation into the direct sum of the quadratic spaces, followed by Euler's identity, makes every component zero. If $k = 1$, the derivatives of a cubic relation lie in the unique quadratic relation; equality of mixed partials forces a pure cube, which cannot occur in a derivative space of quadratic dimension at least nine. If $k = 2$, either the same argument applies or all nonzero cubic

components lie in $\text{Sym}^3 L$ for one two-plane L . A squarefree cubic space contains no nonzero binary cubic, while two remaining components force the common quadratic relation space to contain $\text{Sym}^2 L$, of dimension three, contradicting $k = 2$. Finally, $k \geq 3$ is incompatible with $e + k + c \leq 3$ and $c > 0$. Therefore $c = 0$. $\square$

5. EXCLUSION OF THE 58 STATES

The states are partitioned by mutually exclusive integer conditions. The route count is $1 + 38 + 1 + 2 + 1 + 1 + 5 + 9 = 58$.

TABLE 1. Complete routing of the fixed-six states.

Exclusion mechanism	Number of states
Petersen-triangle equality count at $(s, d, t) = (19, 9, 45)$	1
Global p_d bound and (8)	38
Equality geometry for $d = 9$	1
The bound $p_{11} \leq 55$	2
The bound $p_{12} \leq 80$	1
The sharpened bound $p_{12} \leq 61$	1
Flag-annihilator rank gap for $d = 11, 12$	5
Three-orbit exclusion for $d = 10$	9
Total	58

For direct verification, the complete state list and its unique route assignment are included below. It is generated independently by `scripts/perm5_fixed_six_state_table.py`.

Table 2: The complete fixed-six state universe and its unique exclusion route. The columns p_d^{up} and K_{low} show the displayed global prolongation bound and its substitution in the double-quotient inequality; for $d = 13, 14, 15$ the bound follows from $p_{d+1} \leq p_d + 25$.

s	d	t	h	m_s	$t + d - s$	p_d^{up}	K_{low}	route
19	9	45	54	45	35	35	2160	A
19	9	46	55	45	36	35	2160	B
19	9	47	56	45	37	35	2160	B
19	9	48	57	45	38	35	2160	B
19	9	49	58	45	39	35	2160	B
19	9	50	59	45	40	35	2160	B
19	9	51	60	45	41	35	2160	B
19	10	45	55	45	36	50	2170	B
19	10	46	56	45	37	50	2170	B
19	10	47	57	45	38	50	2170	B
19	10	48	58	45	39	50	2170	B
19	10	49	59	45	40	50	2170	B
19	10	50	60	45	41	50	2170	B
19	11	45	56	45	37	55	2190	B
19	11	46	57	45	38	55	2190	B
19	11	47	58	45	39	55	2190	B

s	d	t	h	m_s	$t + d - s$	p_d^{up}	K_{low}	route
19	11	48	59	45		40	55	2190 B
19	11	49	60	45		41	55	2190 B
19	12	45	57	45		38	61	2209 B
19	12	46	58	45		39	61	2209 B
19	12	47	59	45		40	61	2209 B
19	12	48	60	45		41	61	2209 B
19	13	45	58	45		39	86	2209 B
19	13	46	59	45		40	86	2209 B
19	13	47	60	45		41	86	2209 B
19	14	45	59	45		40	111	2209 B
19	14	46	60	45		41	111	2209 B
19	15	45	60	45		41	136	2209 B
20	9	48	57	48		37	35	2138 B
20	9	49	58	48		38	35	2138 B
20	9	50	59	48		39	35	2138 B
20	9	51	60	48		40	35	2138 B
20	10	48	58	48		38	50	2148 H
20	10	49	59	48		39	50	2148 H
20	10	50	60	48		40	50	2148 H
20	11	48	59	48		39	55	2168 D
20	11	49	60	48		40	55	2168 D
20	12	48	60	48		40	61	2187 E
21	9	48	57	48		36	35	2113 B
21	9	49	58	48		37	35	2113 B
21	9	50	59	48		38	35	2113 B
21	9	51	60	48		39	35	2113 B
21	10	48	58	48		37	50	2123 H
21	10	49	59	48		38	50	2123 H
21	10	50	60	48		39	50	2123 H
21	11	48	59	48		38	55	2143 G
21	11	49	60	48		39	55	2143 G
21	12	48	60	48		39	61	2162 F
22	9	48	57	48		35	35	2088 C
22	9	49	58	48		36	35	2088 B
22	9	50	59	48		37	35	2088 B
22	9	51	60	48		38	35	2088 B
22	10	48	58	48		36	50	2098 H
22	10	49	59	48		37	50	2098 H
22	10	50	60	48		38	50	2098 H
22	11	48	59	48		37	55	2118 G
22	11	49	60	48		38	55	2118 G
22	12	48	60	48		38	61	2137 G

Proposition 5.1 (Non- $d = 10$ states). *Every one of the forty-nine states outside the $d = 10$ row is impossible.*

Proof. The equality cases in the exact shadow computation underlying (6) identify the unique $22 \mapsto 48$ row-star or column-star flag; deleting one cubic basis vector recovers the $21 \mapsto 48$ case by parent counting. These finite claims are reconstructed by the unified replay in Section 8. For $S \subset D$, $G \subset E$, $\partial S \subset G$, and a five-plane $L \subset V$, set

$$W_L = \text{span}\{\partial_\lambda s : s \in S, \lambda \in L^\perp\}.$$

Here a subscript records dimension, so S_{22} and G_{48} denote the complete flag with $\dim S_{22} = 22$ and $\dim G_{48} = 48$. On this flag, $\text{codim}_{G_{48}} W_L \leq 4$; on the deleted-point flag the bound is seven. The failure locus is a projective closed incidence. After Borel degeneration of both the flag and L , the seven row-occupancy partitions of five variables have maximal parent-functional counts

$$4, 4, 2, 1, 1, 0, 0.$$

The independent program streams all $\binom{25}{5} = 53,130$ coordinate five-planes and reproduces the four relevant global maxima 4, 4, 5, 7.

For $1 \leq i \leq 6$, let $L_i := \text{span}\{\ell : \ell \text{ is a factor of } T_i\} \subset V$. Define the addition map and its relation kernel by

$$\mu_F : \bigoplus_{i=1}^6 F_i \longrightarrow U, \quad (f_1, \dots, f_6) \longmapsto \sum_{i=1}^6 f_i, \quad K_F := \ker \mu_F.$$

If pr_i is the i -th coordinate projection, put $R_i := \text{pr}_i(K_F) \subset F_i$. For $u \in U$, choose a lift $(f_1, \dots, f_6) \in \bigoplus_i F_i$ and define $\rho_i(u) := f_i + R_i \in F_i/R_i$. This is independent of the lift. Writing $\rho = (\rho_1, \dots, \rho_6)$, the six-block relation estimate is

$$\rho : U \longrightarrow \bigoplus_i F_i/R_i, \quad \dim \ker \rho \leq 5 \dim K_F.$$

The chain-map identity makes $\rho_i|_G$ vanish on W_{L_i} . The preceding codimension bounds and the strict comparisons, including $43 > 42$ and $48 > 42$, exclude the five relative $d = 11, 12$ states. The Petersen equality count, the exact no-crossing bounds, and the global p_{11}, p_{12} graph bounds exclude the remaining forty-four states. Each finite table is reconstructed from its displayed recurrence or graph and is covered by the unified replay described in Section 8. $\square$

The nine remaining states satisfy

$$s \in \{20, 21, 22\}, \quad d = 10, \quad t \in \{48, 49, 50\}, \quad h = t + 10. \quad (10)$$

Lemma 4.1 gives $H = U$. At least four of the six quadratic blocks have dimension ten. If one such block met E in dimension one, Theorem 3.1 would give $p(W) \leq 26$, whereas (7) gives at least 36. Thus four saturated blocks may be chosen transverse to E .

Choose a twenty-plane in S and apply simultaneous row-column Borel compression to it and to a five-plane of variables. The coordinate limit is a pair

$$\mathcal{S} \subset \binom{[5]}{3} \times \binom{[5]}{3}, \quad \mathcal{L} \subset [5] \times [5],$$

satisfying the following notation. For $\mathcal{A} \subset [5] \times [5]$, define the restricted derivative shadow

$$\partial_{\mathcal{A}} \mathcal{S} = \{(I \setminus \{i\}, J \setminus \{j\}) : (I, J) \in \mathcal{S}, (i, j) \in (I \times J) \cap \mathcal{A}\}.$$

Write $\partial\mathcal{S} := \partial_{[5] \times [5]}\mathcal{S}$ and $\mathcal{L}^c := ([5] \times [5]) \setminus \mathcal{L}$. The coordinate pair satisfies

$$|\mathcal{S}| = 20, \quad |\partial\mathcal{S}| \leq 50, \quad |\partial_{\mathcal{L}^c}\mathcal{S}| \leq 40, \quad |\mathcal{L}| = 5. \quad (11)$$

Simultaneous elementary compression cannot increase either shadow. The layer defect identity reduces (11) to fourteen shifted profiles. Inverse compression and Petersen-fibre analysis divide the profiles into three terminal orbits, conventionally numbered 0, 1, 13.

Proposition 5.2 (Terminal orbit classification). *For a coordinate subset $\mathcal{C} \subset Q$, abbreviate $p(\mathcal{C}) := p(\text{span } \mathcal{C})$. The fourteen shifted profiles and every inverse-compression parent satisfying (11) lie in exactly one of the terminal orbits 0, 1, 13. In orbit 1, the four critical quotients are three length-two local flag quotients and one same-row full-order quotient. Orbit 13 satisfies the uniform structural bound $p(\mathcal{C}) \leq 3|\mathcal{C}|$, with every crossing marginal at most five. Orbit 0 reduces to a Petersen even-cycle identity and five slice formulas.*

Proof. The classification is finite but formula-driven. The orbit-1 endpoint is enumerated by a five-vertex closed formula, ten nonzero Möbius term families, and an eight-row table indexed by the complete graph K_4 on four vertices. The three length-two quotients are excluded by their relative local rings, and the full-order quotient by the same-row valuation calculation. Both are reconstructed by the replay in Section 8. The orbit-13 estimate replaces the historical ten-subset classification. The orbit-0 formulas replace the historical list of ninety-two forest edges. Integer programs independently reconstruct the shifted profiles, visible-shadow parents, fibre profiles, terminal gaps, and the orbit reductions. Their output is compared byte-for-byte with the frozen exact payloads before the result is accepted. $\square$

Proposition 5.3 (Terminal Koszul contradiction). *For every $d = 10$ state, the residual polynomial $R = \text{perm}_5 - (T_1 + \cdots + T_6)$ has Koszul rank at least 2215. A sum of nine quintic Chow terms has Koszul rank at most 2205.*

Proof. In orbit 0, the exact tangent graph has 4100 vertices and kernel dimension eight. Linear reductivity upgrades the trivial cotangent action to the complete local ring. Boolean-Fourier character orthogonality gives the shortening identity and the constancy criterion required by the terminal symbol. Permutations of the three displayed slots generate the full S_4 -symmetry; the resulting prolongation dimensions give $\text{rank } K(R) \geq 2215$. The orbit-1 and orbit-13 terminals satisfy the same or a stronger bound by Proposition 5.2 and their local symbol estimates.

For a single residual Chow term, the relevant quadratic derivative space has dimension at most ten, and its prolongation kernel has dimension at least five. Hence its terminal Koszul rank is at most $25 \cdot 10 - 5 = 245$. Nine terms therefore have total rank at most $9 \cdot 245 = 2205 < 2215$, a contradiction. $\square$

Propositions 5.1 and 5.3 exclude all 58 fixed-six states. Consequently no decomposition of length at most fifteen exists, and

$$\text{ChowRank}(\text{perm}_5) \geq 16. \quad (12)$$

6. THE UPPER BOUND

Glynn's identity [2] specializes to

$$\text{perm}_5 = \frac{1}{16} \sum_{\substack{\epsilon \in \{\pm 1\}^5 \\ \epsilon_0 = 1}} \left(\prod_{r=0}^4 \epsilon_r \right) \prod_{c=0}^4 \left(\sum_{r=0}^4 \epsilon_r x_{rc} \right). \quad (13)$$

The right side has sixteen Chow terms. Walsh cancellation gives coefficient one on the 120 permutation monomials and zero on the other 3005 row-choice monomials. Thus $\text{ChowRank}(\text{perm}_5) \leq 16$, which together with (12) proves Theorem 1.1. The supplied independent expansion checks all 50,000 column choices over the integers. Finally, a decomposition over a characteristic-zero field remains a decomposition over its algebraic closure, while (13) is defined over the prime field. The algebraically closed lower bound and the displayed upper bound therefore give the asserted equality over every characteristic-zero field.

7. COMPUTATIONAL LEMMAS AND CERTIFICATE CONTRACT

This section isolates every finite assertion on which the proof depends. A *certificate contract* means a finite input class, an exact arithmetic domain, an asserted output, and a verifier that terminates with failure unless the asserted output is reproduced. Coordinate variables are ordered by $v(i, j) = 5i + j$, and subsets and pairs are ordered lexicographically. All matrix ranks below are computed by exact row reduction; no numerical tolerance occurs.

Lemma 7.1 (One-intersection endpoint certificate). *For the coordinate fixed points arising in Theorem 3.1, the relative prolongation maxima for four-plane, attached five-plane, and external five-plane flags are, respectively,*

$$22, \quad 26, \quad 22.$$

The exhaustive domain has $21,600 + 432,432 + 432,432 = 886,464$ flags, and exactly twenty attached flags attain the middle maximum.

Certificate proof. For each flag the verifier constructs the matrix of the divided-power cubic prolongation map in the lexicographically ordered coordinate bases. It scans the complete finite domain over each of $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, compares the three histograms, and recomputes every extremal matrix over $\mathbb{Q}$. The implementation in `scripts/perm5_one_intersection_independent_multifield.py` does not import a frozen producer or read its result. Since reduction modulo a prime cannot increase rank, the finite-field kernel dimensions bound their rational counterparts from above. Thus the certificate proves precisely the inequality used in Theorem 3.1. $\square$

Lemma 7.2 (Fixed-six state certificate). *The integer triples (s, d, t) allowed by (7)–(9) comprise exactly fifty-eight states. The eight mutually exclusive routes in Table 1 contain 1, 38, 1, 2, 1, 1, 5, 9 states and exhaust this set.*

Certificate proof. The verifier reconstructs the Petersen-product neighborhood recurrence from the eleven-entry function ν , enumerates the resulting integer triangle, and evaluates the route predicates in their displayed order. It rejects an unrouted state, a multiply routed state, a route count different from the listed vector, or a total different from fifty-eight. The independent entry point is `scripts/perm5_fixed_six_state_table.py`; it also emits the human-readable table included in this article. The predicates are the finite inequalities and equality cases invoked in Propositions 5.1 and 5.2. $\square$

Lemma 7.3 (Parent-table and terminal certificates). *The coordinate five-plane parent maxima used in Proposition 5.1 are 4, 4, 5, 7. The shifted-profile, visible-shadow, and inverse-compression arguments in Proposition 5.3 reduce the terminal calculation to orbit zero, where $\text{rank } K(R) \geq 2215$.*

Certificate proof. The independent parent-table verification streams, rather than stores, all $\binom{25}{5} = 53,130$ coordinate five-planes and recomputes the four maxima. The entry point `scripts/perm5_terminal_independent_verification.py` reconstructs the tangent graph and rational Fourier matrices directly from their definitions and checks the terminal rank gap. The exact finite inputs and outputs are listed in `docs/computational_lemmas.md`. $\square$

Lemma 7.4 (Upper-bound certificate). *The right-hand side of (13) has coefficient one on each of the 120 permutation monomials and coefficient zero on the other 3005 row-choice monomials.*

Certificate proof. The independent integer program `scripts/perm5_glynn_upper_bound_independent.py` expands all $16 \cdot 5^5 = 50,000$ column choices and compares the complete coefficient map with the permanent support. This is a direct verification of the displayed identity, not a probabilistic identity test. $\square$

Lemmas 7.1–7.4 are finite lemmas in the logical proof: their conclusions, not the existence of output files, are used in Sections 3–6.

8. EXACT VERIFICATION

From the repository root, execute

```
python -B perm5/verify_all.py
```

The command requires Python 3 and only its standard library. It performs five independent exact computations:

- (1) the 886,464-flag scan over each of $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, followed by rational checks of all twenty extrema;
- (2) the reconstruction and unique route assignment of all fifty-eight fixed-six states;
- (3) the streamed scan of all $\binom{25}{5} = 53,130$ coordinate five-planes, reproducing the parent-table maxima 4, 4, 5, 7;
- (4) the 4,100-vertex tangent graph and rational Fourier computation, which give $2215 > 2205$; and

(5) the 50,000-choice integer expansion of Glynn's identity.

The map from finite lemmas to programs and exact outputs is given in `docs/computational_lemmas.md`. The verifier fails on a missing file, a changed payload, or a nonzero computation exit. The theorem concerns ordinary Chow rank; it makes no border Chow-rank statement and no equality claim for general n .

REFERENCES

- [1] A. Borel, *Linear algebraic groups*, second ed., Graduate Texts in Mathematics, vol. 126, Springer-Verlag, New York, 1991, doi:10.1007/978-1-4612-0941-6.
- [2] D. G. Glynn, *The permanent of a square matrix*, European J. Combin. **31** (2010), no. 7, 1887–1891, doi:10.1016/j.ejc.2010.01.010.
- [3] N. Ilten and Z. Teitler, *Product ranks of the 3×3 determinant and permanent*, Canad. Math. Bull. **59** (2016), no. 2, 311–319, doi:10.4153/CMB-2015-076-1.
- [4] G. O. H. Katona, *A theorem of finite sets*, in: Theory of Graphs (Proc. Colloq., Tihany, 1966) (P. Erdős and G. O. H. Katona, eds.), Academic Press, New York; Akadémiai Kiadó, Budapest, 1968, pp. 187–207.
- [5] J. B. Kruskal, *The number of simplices in a complex*, in: Mathematical Optimization Techniques (R. Bellman, ed.), University of California Press, Berkeley, 1963, pp. 251–278, doi:10.1525/9780520319875-014.